

Simple critical zeros in short intervals: stability, parity, localization, Hilbert compression, and spectral defect

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CAOS Research program

github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Let N , S , and O count, respectively, all copies, simple critical zeros, and distinct odd-multiplicity critical zeros in a fixed power interval. Let Q be the squared-kernel pair sum and $D(G)$ the convex spectral defect of the simple-real Gram matrix. We prove the sharp finite defect-parity inequality

$$(Q - S - D(G))(N - O) \geq 2(N - S)^2.$$

We also prove that Pearce-Crump's vector-profile Selberg detector localizes to every fixed finite rank. Applying this rank-independent transfer to the source-certified six-square constant C_6 and then using the Hilbert-parity product gives

$$\liminf \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2k_6(\theta)}{3}, h_6(\theta) \right\},$$

where $k_6(\theta) = (\theta - 1/2)/(4eC_6)$ and

$$h_6(\theta) = \frac{3 + k_6(\theta) - \sqrt{(1 - k_6(\theta))(9 - k_6(\theta) - 8c(\theta))}}{4}.$$

Exact rational enclosures place the unique positivity onset in $(0.5458837, 0.5458838)$. At $\theta = 0.545884$, where the rank-three term is negative, $h_6 > 2.5541123454 \dots \times 10^{-7}$; at $\theta = 0.5459$, $h_6 > 0.0000177645181613023236$. A radius-optimized pressure transfer further gives a spectral term $H_6 > h_6$ throughout the positive curve. The finite coefficient two is sharp. The six-square profile and constant are attributed to the source; its coefficient matrix is not printed and is not independently reconstructed here. The results are asymptotic and depend on recent preprints. No global percentage record, effective starting height, external peer review, or solution of the Riemann hypothesis is claimed.

Summary of results

The central finite result retains both the simple-real Gram spectral defect and Lamzouri’s first Hilbert-subspace dimension, then uses parity to compare the dimension with distinct odd support. The new analytic result shows that the local Selberg transfer is valid for every fixed finite-rank vector profile. Using the smaller source-certified six-square constant moves the explicit positivity onset from the rank-three interval $(0.5458846, 0.5458847)$ to $(0.5458837, 0.5458838)$.

The spectral-defect transfer remains a secondary strict improvement of the complete positive curve. Its rational radius ratio $11/5$ gives a larger correlated certificate than the earlier transparent choice 4. Further results are the direct finite stability transfer, the odd-frame pressure theorem, its stronger consequence throughout the positive short-interval curve, and a certified two-variable pressure bound. The earlier classical parity theorem remains a separate weaker-input qualitative result.

Verification. For the pressure bounds, parity deduction, and local Selberg, Hilbert–parity, and defect–parity transfers (computations EXP-003 through EXP-008 [20, 21, 22, 23, 24, 25]), the verification consists of exact finite checks, complete 256-bit replays of the interval certificates, checks of the imported statements against their primary sources, and independent step-by-step checks of the proofs, including an independent recomputation of every census row. It is not external peer review or end-to-end formal verification.

Interval-range quantity	Certified value
Wang cosine positivity threshold	0.550193964744154...
Localized Selberg–parity threshold	(0.5459, 0.546)
Rank-three Hilbert–parity onset	(0.5458846, 0.5458847)
Rank-six Hilbert–parity onset	(0.5458837, 0.5458838)
Rank-six lower bound at $\theta = 0.545884$	$> 2.5541123454 \times 10^{-7}$
Rank-six lower bound at $\theta = 0.5459$	> 0.0000177645181613
Optimized spectral gain at $\theta = 0.5459$	$> 1.7766622541 \times 10^{-68}$

Bound at $\theta = 3/4$	Simple-critical proportion	Gain above Wang
Wang	0.4190750129754...	0
Direct transfer, Cor. 1.8	0.4190768284253...	$1.8154498796 \times 10^{-6}$
Odd-frame bound, Cor. 1.9	0.4190777360205...	$2.7230451445 \times 10^{-6}$
Two-variable pressure certificate, Cor. 1.9	0.4190878881701...	$1.2875194687 \times 10^{-5}$

1 Statement and relation to prior work

Let $N(T, H)$ count the nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $T < \gamma \leq T + H$, including multiplicity. Let $N_0^s(T, H)$ count the simple zeros among them with $\beta = 1/2$, and let $N^d(T, H)$ count distinct complex zeros, each once. Let $N_{\text{odd}}(T, H)$ count distinct critical zeros of odd multiplicity, also each once. It does not count their multiplicity. Every proportion below has denominator $N(T, H)$.

The recent argument discovered by Claude and communicated by Alpöge and Furman [1] made the optimized Montgomery–Taylor simple-critical proportion $0.672500703679 \dots$ unconditional. Lamzouri [2] replaced its finite-compression machinery by a Hilbert-space inequality for conjugation-invariant multisets. Wang [5] established the needed short-interval pair-correlation

theorem and deduced, for fixed $0 < \theta < 1$,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c(\theta), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c(\theta)}{2}, \quad (1)$$

where

$$c(\theta) = 2 - \frac{\theta}{2} - \frac{1}{\sqrt{2}} \cot(\theta/\sqrt{2}). \quad (2)$$

The derivative is $c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) > 0$. Since $c(\theta) \rightarrow -\infty$ as $\theta \downarrow 0$ and $c(1) > 0$, there is a unique zero $\theta_0 \in (0, 1)$, numerically $\theta_0 \approx 0.550193964744154\dots$, as printed by Wang. The proof below uses its defining equation and rational comparisons, not a numerical root certificate.

The stable rank–trace inequality, convex Gram defect, pinching method, and cosine-root non-additivity used here are prior work in the `ainta` repository [13]. Global pressure windows, larger frames, nonuniform pair capacities, spectral envelopes, and combinations of certificates are developed in [14, 16, 17]. The alternating schedule used below is a simple choice within this established framework. We do not claim these ingredients or an improved global record. Our contributions are the parity-compressed product inequality, its quadratic short-interval transfer, the explicit threshold below 0.545885, the direct finite-stability transfer, and the certified pressure refinements. Classical odd-multiplicity density and retained multiplicity excess extend the range qualitatively. Pearce-Crump’s coefficient-uniform detector localizes to every fixed power interval above one half and supplies the explicit odd-support curve. The new product retains Lamzouri’s attributed first-subspace dimension before scalar compression; its coefficient inequality itself is not claimed as new. All finite proofs keep off-line signs, multiplicities, and singular Gram matrices explicit.

Selberg’s positive density theorem at lengths $T^{1/2+\epsilon}$ is reproduced as Theorem B in Karatsuba’s primary article [9]. Karatsuba’s own theorem works at $T^{27/82+\epsilon}$. These count distinct odd-order critical zeros. They do not already assert a positive simple-zero proportion. Steuding’s later simple-critical theorem gives the explicit exponent 0.552 [10, 11]; Wang gives the smaller explicit threshold θ_0 . Pearce-Crump [6] later gave a coefficient-uniform sign detector with a certified diagonal constant. Its global theorem exceeds seven percent on the critical line. The localization below uses the paper’s arbitrary-subinterval off-diagonal estimate; it does not alter the detector or claim the global percentage as a new result.

The labels [D] and [MV] distinguish mathematical deductions from finite machine-verifiable statements. The following main range result is a deduction from the stated primary analytic inputs. Exact finite checks verify its algebra; they do not replace those inputs or prove all-height statements by enumeration.

Theorem 1.1 (Finite-rank local Selberg transfer, D+MV). *Let q be any fixed finite-rank admissible vector profile covered by Pearce-Crump’s vector-profile theorem, with diagonal constant $C[q]$. For every fixed $1/2 < \theta < 1$,*

$$\liminf_{T \rightarrow \infty} \frac{N_{\text{odd}}(T, T^\theta)}{N(T, T^\theta)} \geq \frac{\theta - 1/2}{4eC[q]}. \quad (3)$$

For the fully printed rank-three profile,

$$C_3 \in 0.6567752140190419405677628751089899133 \pm 10^{-17}.$$

The source also states that a fixed ten-direction, six-square profile has

$$C_6 \in 0.6566338678379319741683641732 \pm 5.63 \times 10^{-18}. \quad (4)$$

Its coefficient matrix is not printed, so (4) is an attributed source-certified input rather than an independently reconstructed certificate. With

$$k_6(\theta) = \frac{\theta - 1/2}{4eC_6},$$

one obtains

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2k_6(\theta)}{3} \right\}. \quad (5)$$

Theorem 1.2 (Rank-six Hilbert-parity compression, D+MV). *Put*

$$h_6(\theta) = \frac{3 + k_6(\theta) - \sqrt{(1 - k_6(\theta))(9 - k_6(\theta) - 8c(\theta))}}{4}.$$

Then

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2k_6(\theta)}{3}, h_6(\theta) \right\}. \quad (6)$$

The new term has a unique zero θ_6 and

$$0.5458837 < \theta_6 < 0.5458838. \quad (7)$$

For comparison, the corresponding rank-three root lies in $(0.5458846, 0.5458847)$. At $\theta = 0.545884$ the rank-three term is negative, while

$$h_6(0.545884) > 2.5541123454645702 \times 10^{-7}. \quad (8)$$

At $\theta = 0.5459$,

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^{0.5459})}{N(T, T^{0.5459})} > 0.0000177645181613023236390595079. \quad (9)$$

The improvement over the rank-three term exceeds $9.2635430617773560 \times 10^{-7}$.

Theorem 1.3 (Spectral-defect parity improvement, D+MV). *Fix $1/2 < \theta < 1$ with $h = h_6(\theta) > 0$. In (18), take $R = 11/(5h)$ and write*

$$d_\theta = (b/B)^2, \quad \alpha_\theta = \frac{2d_\theta}{5}, \quad \beta_\theta = \frac{4d_\theta h}{11}, \quad \kappa_\theta = k_6(\theta), \quad q_\theta = 2 - c(\theta).$$

Let

$$H_6(\theta) = \frac{4 - (1 + \alpha_\theta)(1 - \kappa_\theta) - \sqrt{[4 - (1 + \alpha_\theta)(1 - \kappa_\theta)]^2 - 8[2 - (1 - \kappa_\theta)(q_\theta + \beta_\theta)]}}{4}. \quad (10)$$

Then

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2k_6(\theta)}{3}, H_6(\theta) \right\}, \quad (11)$$

and

$$H_6(\theta) - h_6(\theta) \geq \frac{(1 - k_6(\theta))(\alpha_\theta h_6(\theta) - \beta_\theta)}{4} > 0. \quad (12)$$

At $\theta = 0.5459$, exact directed rational arithmetic proves

$$H_6(0.5459) - h_6(0.5459) > 1.7766622541125682 \times 10^{-68}. \quad (13)$$

The spectral correction does not change the onset in (7).

Theorem 1.1 refines the simple-critical range numerically for $\theta > 1/2$. The next theorem uses only a fixed classical sign-change seed and remains a separate qualitative statement under weaker input.

Theorem 1.4 (Parity transfer and qualitative range extension, D). *Put $\alpha = 51/100$. There exists a constant $\kappa > 0$ such that, for every fixed $\alpha < \theta < 1$,*

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ 0, c(\theta), \frac{c(\theta) + 2\kappa}{3} \right\}, \quad (14)$$

$$\liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \max \left\{ \kappa, \frac{1 + c(\theta)}{2}, \frac{3 + 2c(\theta) + \kappa}{6} \right\}. \quad (15)$$

One fixed κ works for all these fixed exponents; the eventual height may depend on θ . Define

$$\delta = \min \left\{ \frac{\theta_0 - \alpha}{2}, \frac{\kappa}{4} \right\} > 0, \quad \theta_1 = \theta_0 - \delta. \quad (16)$$

Then $\alpha < \theta_1 < \theta_0$ and the lower simple-critical proportion at every fixed $\theta_1 \leq \theta < 1$ is at least $\kappa/3 > 0$. Separately, set

$$\delta_{\text{half}} = \min \left\{ \frac{\theta_0 - \alpha}{2}, \frac{\kappa}{16} \right\}, \quad \theta_{\text{half}} = \theta_0 - \delta_{\text{half}}. \quad (17)$$

For every fixed $\theta_{\text{half}} \leq \theta < 1$, the lower distinct-zero proportion is at least $1/2 + \kappa/12$.

The distinct claim is greater than one half on some shorter intervals; it is not a new exponent for merely positive distinct density, which already follows from the classical odd-zero theorem. Theorem 1.4 does not give an effective value of κ , a numerical value of either new exponent, or a uniform theorem when θ varies with T .

Theorem 1.5 (Explicit three-point refinement, D). *Fix $\theta_0 < \theta < 1$ and $R > 2/c(\theta)$. Define*

$$\begin{aligned} a &= \theta/\sqrt{2}, & q &= a \tan a, & M &= \pi\theta R, \\ B &= q(1 + M^2/a^2), \\ b &= \min \left\{ \frac{q}{2}, \frac{q^4}{4(q + M)^2(M + 3q/2)} \right\}, & \delta &= \frac{1}{2} \min\{1, 2(b/B)^2\}. \end{aligned} \quad (18)$$

Then $0 < \delta \leq 1/2$, and, with

$$c_*(\theta, R) = c(\theta) + \frac{\delta\{c(\theta) - 2/R\}}{3 - \delta} > c(\theta), \quad (19)$$

we have

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)} \geq c_*(\theta, R), \quad \liminf_{T \rightarrow \infty} \frac{N^d(T, T^\theta)}{N(T, T^\theta)} \geq \frac{1 + c_*(\theta, R)}{2}. \quad (20)$$

In particular, $R = 4/c(\theta)$ gives a fully specified positive improvement.

Theorem 1.6 (Odd-frame pressure refinement, D). *Fix $\theta_0 < \theta < 1$. Suppose $p, \varepsilon > 0$ and the cosine kernel K_θ of (47) satisfies*

$$2\{K_\theta(u)^2 + K_\theta(v)^2 + K_\theta(u+v)^2\} + p(u+v) \geq \varepsilon \quad (u, v \geq 0). \quad (21)$$

For every fixed integer $k \geq 1$ with $k\varepsilon \leq 1$, put

$$c_{\text{odd}} = c(\theta) + \frac{k\{\varepsilon c(\theta) - 2p\}}{2k + 1 - k\varepsilon}. \quad (22)$$

Then the lower asymptotic simple-critical and distinct proportions are at least c_{odd} and $(1+c_{\text{odd}})/2$, respectively. The gain is strictly positive when $\varepsilon c(\theta) > 2p$.

Corollary 1.7 (Stronger entire positive curve, D). *For the analytic δ in (18), one has $0 < \delta < 1/4$. Every integer $k > 1$ with $k\delta \leq 1$ strengthens Theorem 1.5 by replacing its gain coefficient $\delta/(3-\delta)$ with $k\delta/(2k+1-k\delta)$. In particular $k=2$ is available for every fixed $\theta_0 < \theta < 1$.*

The conservative gap in Theorem 1.5 requires no numerical search. A stronger finite certificate gives the following example.

Corollary 1.8 (Compact-certificate example, D+MV). *Set $\theta = 3/4$, $R = 21/4$, $d = 1/7000$, and*

$$C_* := \frac{3c(3/4) - 2d/R}{3 - d} = 0.419076828425303996736665787527 \dots$$

Then the simple-critical proportion in (20) is at least C_ , and the distinct-zero proportion is at least*

$$\frac{1 + C_*}{2} = 0.709538414212651998368332893763 \dots$$

The improvement in the simple-critical proportion is $0.000001815449879663002112176828 \dots$

Corollary 1.9 (Odd-frame bound and two-variable pressure certificate, D+MV). *At $\theta = 3/4$, the compact energy certificate of Section 10, with $d = 1/7000$ and $R = 21/4$, gives, with $k = 7000$,*

$$C_A = c(3/4) + \frac{c(3/4) - 8/21}{14000} = 0.419077736020568836833221725071 \dots \quad (23)$$

The all-gap pressure certificate of Appendix B has

$$p = \frac{1}{12500}, \quad \varepsilon = \frac{443239}{10^9}, \quad k = 2256, \quad 2k + 1 = 4513. \quad (24)$$

It yields

$$\begin{aligned} C_B &= c(3/4) + \frac{2256\{(443239/10^9)c(3/4) - 2/12500\}}{4513 - 2256(443239/10^9)} \\ &= 0.419087888170111727959091183775 \dots \end{aligned} \quad (25)$$

The respective distinct companions are

$$(1 + C_A)/2 = 0.709538868010284418416610862535 \dots,$$

$$(1 + C_B)/2 = 0.709543944085055863979545591887 \dots$$

The conclusion is for each fixed exponent and all sufficiently large heights in the asymptotic sense. We do not provide an effective starting height or uniformity as an exponent approaches its endpoint. The pressure corollaries themselves concern $\theta > \theta_0$; the distinct parity argument in Theorem 1.4 supplies the qualitative extension below θ_0 . The common analytic dependency is Wang's inspected preprint theorem, whose underlying pair-correlation method is due to Baluyot, Goldston, Suriajaya, and Turnage-Butterbaugh [15, 18].

2 A stable inequality for finite multisets

Let Z be a finite conjugation-invariant multiset in $\mathbb{C}$. We sum over its distinct support with positive integer multiplicities $m_z = m_{\bar{z}}$, and put $N = \sum_z m_z$. Choose a real even compactly supported $\eta \in L^2(\mathbb{R})$ with $\int \eta^2 = 1$, and define

$$v_z(u) = \eta(u)e^{-2\pi izu}, \quad K(t) = \widehat{\eta^2}(t), \quad \widehat{f}(t) = \int_{\mathbb{R}} f(u)e^{-2\pi itu} du.$$

Every v_z belongs to L^2 because Z is finite and the support is compact. Work in the finite-dimensional span $\mathcal{H}$ of these vectors. Our inner product is $\langle v, w \rangle = \int \bar{v}w$, linear in the second variable; write $|v\rangle\langle w|$ for $h \mapsto v\langle w, h \rangle$.

Define the reflected rank-one operator

$$A = \sum_z m_z |v_z\rangle\langle v_{\bar{z}}|. \quad (26)$$

It is selfadjoint by conjugation invariance. Direct multiplication gives

$$\mathrm{tr} A = N, \quad Q := \|A\|_{\mathrm{HS}}^2 = \mathrm{tr} A^2 = \sum_{z,w} m_z m_w K(z-w)^2. \quad (27)$$

Indeed, $\langle v_{\bar{z}}, v_z \rangle = \int \eta^2 = 1$, while the trace of the product of the z and w terms, before multiplicity factors, is $K(w-z)K(z-w) = K(z-w)^2$, since K is even. The ordinary square in (27) is essential. Individual terms for complex arguments need not be positive or even real; their complete sum is a nonnegative operator norm.

Let $x_1 < \dots < x_s$ be the simple real points of Z . Let r be the number of distinct multiple real points and k the number of distinct nonreal conjugate pairs. Put

$$P = \sum_{j=1}^s |v_{x_j}\rangle\langle v_{x_j}| = VV^*, \quad G = V^*V = (K(x_i - x_j))_{i,j=1}^s, \quad C = A - P.$$

Then G is positive semidefinite with diagonal one and $\mathrm{tr} P = s$. For each nonreal pair put $g = (v_z + v_{\bar{z}})/2$ and $h = (v_z - v_{\bar{z}})/(2i)$. Its contribution to C is $2m_z(|g\rangle\langle g| - |h\rangle\langle h|)$. The form of C is nonpositive on the orthogonal complement of the r multiple-real vectors and k vectors g . Thus

$$n_+(C) \leq r + k, \quad N \geq s + 2r + 2k. \quad (28)$$

No distance of a nonreal point from the real axis is assumed.

Lemma 2.1 (Known stable rank–trace inequality). *Suppose $P = VV^*$ on a finite-dimensional Hilbert space, the s columns of V have norm one, C is selfadjoint, and $n_+(C) \leq d$. Define*

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2. \end{cases}$$

Then

$$\|P + C\|_{\mathrm{HS}}^2 \geq 4\mathrm{tr}(P + C) - 3s - 4d + \mathrm{tr} \Psi(V^*V). \quad (29)$$

Proof. This is the unit-column case of the stability inequality in [13, Section 2]. We give a spectral proof that keeps all zero eigenvalues. Let $m = \dim \mathcal{H}$ and let $p_1 \leq \dots \leq p_m$ be the eigenvalues of

P . If $d < m$, the min-max principle gives $\lambda_i(P + C) \leq p_{i+d}$ for $1 \leq i \leq m - d$: intersect the span of the first $i + d$ eigenvectors of P with a codimension-at-most- d subspace on which $C \leq 0$.

It follows that

$$\operatorname{tr}(P + C - 2I)^2 \geq \sum_{j=d+1}^m (2 - p_j)_+^2 \geq \sum_{j=1}^m (2 - p_j)_+^2 - 4d.$$

Here $0 \leq (2 - p_j)_+^2 \leq 4$. If $d \geq m$, the final right-hand side is nonpositive and the same inequality is immediate. For every $p \geq 0$, $(2 - p)_+^2 = \Psi(p) - 2p + 3$. The nonzero spectra of VV^* and V^*V agree, and $\Psi(0) = 1$, so

$$\operatorname{tr} \Psi(P) = \operatorname{tr} \Psi(V^*V) + m - s.$$

Substitute these identities and $\operatorname{tr} P = s$, and expand the left-hand square. All dimensional terms cancel and give (29). $\square$

Proposition 2.2 (Finite Hilbert stability transfer). *For the multiset and operator above, write $D(G) = \operatorname{tr} \Psi(G)$ and $D_Z = s + r + 2k$ for the number of distinct support points. Then*

$$s \geq 2N - Q + D(G), \quad D_Z \geq \frac{3N - Q + D(G)}{2}. \quad (30)$$

Proof. By Lemma 2.1 and (28), $Q \geq 4N - 3s - 4(r + k) + D(G)$. The multiplicity bound in (28) makes this at least $2N - s + D(G)$. For the second claim, the difference between $4N - 3s - 4(r + k)$ and $3N - 2D_Z$ is $N - s - 2r \geq 0$. $\square$

This proof retains the signed off-line remainder throughout. The positive Gram defect belongs only to the actual simple real points. The second inequality in (30), rather than elementary counting from a simple-zero proportion, will yield the distinct-zero companion.

2.1 Retaining multiplicity excess and parity

Keep the notation of Proposition 2.2; in this subsection k counts nonreal pairs, not frame length. Let O count the distinct real support points of odd multiplicity, and define the nonnegative excess

$$E = N - s - 2r - 2k. \quad (31)$$

An atom of real multiplicity $m \geq 2$ contributes $m - 2$ to E ; a nonreal pair of multiplicity m contributes $2(m - 1)$. Thus

$$O \leq s + E. \quad (32)$$

Indeed, a nonsimple odd real point has multiplicity at least three and contributes at least one to E . This is a statement about distinct odd points, not a bound for odd-multiplicity copies.

Proposition 2.3 (Exact parity and distinct residuals). *For the finite multiset above,*

$$3s \geq 2N - Q + 2O + D(G), \quad 6D_Z \geq 7N - 2Q + O + 2D(G). \quad (33)$$

Proof. The strong form already proved in Proposition 2.2 is

$$Q \geq 4N - 3s - 4r - 4k + D(G) = 2N - s + 2E + D(G).$$

Write its nonnegative slack as $\sigma = Q - (4N - 3s - 4r - 4k + D(G))$. Direct substitution gives the exact identities

$$3s - \{2N - Q + 2O + D(G)\} = 2(s + E - O) + \sigma, \quad (34)$$

$$6D_Z - \{7N - 2Q + O + 2D(G)\} = (s + E - O) + 6k + 2\sigma. \quad (35)$$

Both right-hand sides are nonnegative by (32). $\square$

For clarity, the atomic contributions to $s + E - O$ are zero at a simple real point, $m - 2$ at a real even point, $m - 3$ at a real odd point of multiplicity at least three, and $2(m - 1)$ at a nonreal pair. This proves the bookkeeping for all integer multiplicities without a finite cutoff. The parity inequalities are consequences of the known stable multiplicity bound, not a claim of a new rank–trace theorem.

Remark 2.4 (Sharp accounting and a false shortcut). The inequality $2D_Z \geq N + s$ is false for general multisets: one real point of multiplicity three has $(N, s, D_Z) = (3, 0, 1)$. The exact difference is $2D_Z - N - s = 2k - E$. Thus one cannot take one half of one plus the parity simple-zero bound to obtain (15).

Both residuals can vanish in an actual finite Hilbert model. Take the probability density $\mathbf{1}_{[-1/2, 1/2]}$ and distinct integer support points with multiplicities one or two. Their features are orthonormal, so $Q = s + 4r$, $N = s + 2r$, $O = s$, and $E = k = D(G) = \sigma = 0$. This is sharpness of the finite bookkeeping. It does not assert that zeta zeros realize these configurations.

One can also express the algebra as a scalar optimization. From normalized bounds $s - 2e \geq c$ and $s + e \geq o$, with $s, e, o \geq 0$, the exact minimum of s in this relaxation is

$$\max\{0, c, (c + 2o)/3\}. \quad (36)$$

The three terms are immediate lower bounds. Conversely, take s equal to the displayed maximum and $e = \max\{0, o - s\}$. If $s \geq o$, then $e = 0$ and $s \geq c$; otherwise $s - 2e = 3s - 2o \geq c$. This proves feasibility and the stated relaxation optimum. Extra total-mass or Gram constraints are not included, so no optimum over actual zero configurations is claimed.

2.2 Hilbert-dimension parity compression

The next proposition returns to Lamzouri’s Hilbert construction rather than the stable inequality above. Its arbitrary-parameter coefficient estimate is an attributed input [2, 1]; the parity-compressed product is the new deduction.

Proposition 2.5 (Sharp Hilbert–parity product). *For a nonempty conjugation-invariant finite multiset, let N count copies, S count simple real support points, and O count distinct real support points of odd multiplicity. With the squared-kernel pair sum Q defined in (27),*

$$(Q - S)(N - O) \geq 2(N - S)^2. \quad (37)$$

The coefficient two is sharp.

Proof. List the nonsimple real points and nonreal conjugate pairs as in Lamzouri’s proof of Proposition 2.1 [2]. The first Gram–Schmidt range has exact dimension $d = r + k$ by linear independence of distinct exponentials on the positive-measure set where the kernel factor is nonzero. If A_U, A_M, A_L are the coefficient sums in the first, middle, and last ranges, that proof gives

$$A_U + A_M + A_L = N, \quad A_M \leq S, \quad \alpha_j \leq 0 \text{ in the last range.}$$

Bessel's inequality and, for any $t \geq 1$, the scalar bounds $x^2 \geq 2tx - t^2$, $x^2 \geq 2x - 1$, and $x^2 \geq 2tx$ on the three ranges give

$$Q \geq 2tN - (2t - 1)S - t^2d. \quad (38)$$

This is the attributed arbitrary-parameter inequality. If $d = 0$, then $N = S = O$ and (37) is immediate. Otherwise multiplicity gives $N - S \geq 2d$, so $t = (N - S)/d \geq 2$ is admissible. Substitution in (38) yields

$$Q \geq S + \frac{(N - S)^2}{d}. \quad (39)$$

If $o = O - S$ counts odd nonsimple real support points, then each such point costs one copy beyond the two-copy floor. Hence

$$N - S \geq 2r + o + 2k = 2d + O - S, \quad 2d \leq N - O.$$

Combining this with (39) proves (37). A single real point of multiplicity two has $(N, S, O, Q) = (2, 0, 0, 4)$, and multiplicity three has $(3, 0, 1, 9)$; both give equality. $\square$

Proposition 2.6 (Sharp spectral-defect parity product). *In the setting of Proposition 2.5, let G be the Gram matrix of the S simple real vectors and $D(G) = \text{tr } \Psi(G)$, with Ψ from Lemma 2.1. Then*

$$(Q - S - D(G))(N - O) \geq 2(N - S)^2. \quad (40)$$

The coefficient two is sharp.

Proof. The arbitrary-parameter multiplicity-aware rank-trace theorem in the attributed formal source [3] uses

$$g_t(x) = x^2 - tx - (x - t)_+^2.$$

Apply it to the positive operator formed by the simple real vectors and the selfadjoint remainder, whose positive index is at most $d = r + k$. Evaluating the positive operator in an eigenbasis gives

$$Q \geq tS + \sum_{j=1}^S g_t(\lambda_j) + 2t(N - S) - t^2d, \quad (41)$$

where all S eigenvalues of G , including zeros, are retained. For $t \geq 2$ define

$$\Psi_t(x) = g_t(x) + (t - 2)x + 1 = \begin{cases} (x - 1)^2, & 0 \leq x \leq t, \\ (t - 1)(2x - t - 1), & x \geq t. \end{cases}$$

Since $\sum_j \lambda_j = S$, equation (41) becomes

$$Q \geq 2tN - (2t - 1)S - t^2d + D_t(G), \quad D_t(G) = \text{tr } \Psi_t(G). \quad (42)$$

The $t = 2$ eigenbasis interpretation is also recorded in the inspected teal-sea bridge [4]. Direct comparison of the three ranges $0 \leq x \leq 2$, $2 \leq x \leq t$, and $x \geq t$ gives $\Psi_t(x) \geq \Psi(x)$. Hence $D_t(G) \geq D(G)$.

When $d > 0$, substitute $t = (N - S)/d \geq 2$ in (42) to obtain

$$Q \geq S + \frac{(N - S)^2}{d} + D(G).$$

Together with $2d \leq N - O$, proved in Proposition 2.5, this is (40). The case $d = 0$ is immediate. The double and triple one-point examples have $S = D(G) = 0$ and retain equality, so the coefficient cannot increase. $\square$

3 From three-point energy to a density gain

For a positive semidefinite matrix M with diagonal one, define $\mathcal{E}(M) = 2 \sum_{i < j} |M_{ij}|^2$.

Lemma 3.1 (Block defect and pinching). *We have $D(M) \geq \min\{1, \mathcal{E}(M)\}$. If M_j are the principal blocks of a partition of M , then $D(M) \geq \sum_j D(M_j)$.*

Proof. If all eigenvalues of M are at most 2, then $D(M) = \text{tr}(M - I)^2 = \mathcal{E}(M)$. If some eigenvalue exceeds 2, its own Ψ value exceeds 1 and all other terms are nonnegative. For pinching, diagonalize each principal block and let (w_i) be the resulting orthonormal basis of the full space. Scalar Jensen applied to the spectral resolution of M gives

$$\sum_j D(M_j) = \sum_i \Psi(\langle w_i, M w_i \rangle) \leq \sum_i \langle w_i, \Psi(M) w_i \rangle = D(M).$$

Only scalar convexity of Ψ is used; no operator-convex assertion is needed. $\square$

Lemma 3.2 (Three shifted partitions). *Suppose $0 < d \leq 1$, $R > 0$, and the real even kernel satisfies*

$$2\{K(u)^2 + K(v)^2 + K(u+v)^2\} \geq d, \quad u, v \geq 0, \quad u+v \leq R. \quad (43)$$

If $\ell = x_s - x_1$ is the span of the simple real points, with $\ell = 0$ when $s \leq 1$, then

$$D(G) \geq \frac{d}{3} \left(s - 2 - \frac{2\ell}{R} \right). \quad (44)$$

Proof. For $s \geq 3$, the sum of the spans of its $s - 2$ consecutive triples is at most 2ℓ , since each successive gap occurs at most twice. Hence at most $2\ell/R$ triples have span greater than R . Every remaining triple has defect at least d by Lemma 3.1.

Separate the starting indices of consecutive triples into their three residue classes modulo 3. Within each class the triples are disjoint principal blocks; add the unused points as singleton blocks. Apply pinching to each partition and average the three inequalities. Every short triple has been counted once, proving (44). If its right-hand side is negative, nonnegativity proves the inequality; this also covers $s < 3$. $\square$

The factor $1/3$ records the three shifted partitions. Overlapping triples cannot simply be summed without this step.

3.1 Odd frames and exact pressure accounting

The next construction adds energies of pair-disjoint triples inside one frame. It applies defect pinching only between disjoint full frames. This distinction is essential: arbitrary overlapping triple defects cannot be summed. For instance, two consecutive three-by-three all-ones blocks have total defect ten, while their four-by-four all-ones parent has defect eight.

Lemma 3.3 (Odd-frame pressure bound). *Let K be a real even positive-type kernel with $K(0) = 1$, satisfying (21) with parameters $p, \varepsilon > 0$. Let $k \geq 1$, $M = 2k + 1$, and $k\varepsilon \leq 1$. For the Gram matrix of s ordered points of total span ℓ , set*

$$\alpha = \frac{k\varepsilon}{M}, \quad \beta = \frac{(M-1)p}{M} = \frac{2kp}{M}.$$

Then

$$D(G) \geq \alpha(s - M + 1) - \beta\ell. \quad (45)$$

As before, set $\ell = 0$ for $s \leq 1$.

Proof. For an M -point frame $y_1 \leq \dots \leq y_M$, select triples $(y_{2j-1}, y_{2j}, y_{2j+1})$, $1 \leq j \leq k$. They share at most one vertex and no unordered pair. Thus their doubled pair energies sum to at most $\mathcal{E}(G_F)$, while their spans telescope:

$$\sum_{j=1}^k (y_{2j+1} - y_{2j-1}) = y_M - y_1 =: \ell_F.$$

Adding their pressure inequalities gives $\mathcal{E}(G_F) + p\ell_F \geq k\varepsilon$. By Lemma 3.1,

$$D(G_F) + p\ell_F \geq k\varepsilon. \quad (46)$$

Indeed, if $\mathcal{E}(G_F) \leq 1$, then $D(G_F) \geq \mathcal{E}(G_F)$; otherwise $D(G_F) \geq 1 \geq k\varepsilon$. This is where the unit cap is used.

Partition the full sequence into complete frames with starts in one residue class modulo M , adding unused indices as singletons. Within each class the frames are disjoint. Across all M classes, there are exactly $(s - M + 1)_+$ complete frames, one for each possible start. Expanding their spans into adjacent gaps gives

$$\sum_{a=1}^{s-M+1} (x_{a+M-1} - x_a) \leq (M-1)\ell,$$

because each gap occurs at most $M-1$ times. Apply pinching and (46) in each class and average. This gives (45) for $s \geq M$. For $s < M$ its right side is nonpositive, so nonnegativity of D completes the proof. $\square$

The construction is a simple nonuniform pair-capacity schedule within the prior global pressure/frame methods of [13, 14, 16, 17]. Its short-interval consequence is obtained below from the exact finite operator, without an additional large-frame Gram approximation.

4 A quantitative obstruction for the cosine kernel

For $0 < \lambda \leq 1$ put

$$f_\lambda(u) = \frac{\cos(\sqrt{2}u)}{\sqrt{2}\sin(\lambda/\sqrt{2})} \mathbf{1}_{[-\lambda/2, \lambda/2]}(u), \quad K_\lambda = \widehat{f}_\lambda. \quad (47)$$

This is a nonnegative probability density. Set $a = \lambda/\sqrt{2}$, $q = a \tan a$, and $X = \pi\lambda t$. Integration yields

$$K_\lambda(t) = \frac{\cos X - (X/a) \cot(a) \sin X}{1 - X^2/a^2}. \quad (48)$$

The apparent singularities at $X = \pm a$ are removable. Equivalently, with $\text{sinc}(x) = \sin(x)/x$ continuously extended at zero,

$$K_\lambda(t) = \frac{\text{sinc}(X - a) + \text{sinc}(X + a)}{2 \text{sinc}(a)}. \quad (49)$$

An actual positive root satisfies $X \tan X = q$. If $X, Y, X+Y$ all corresponded to positive roots, tangent addition would force $X^2 + XY + Y^2 + q^2 = 0$. The following estimate gives a quantitative version without dividing by a possibly vanishing cosine or removable denominator.

Lemma 4.1 (Explicit positive triangle energy). *Fix $R > 0$, put $M = \pi\lambda R$, and define B, b as in (18), using $a = \lambda/\sqrt{2}$ and $q = a \tan a$. Then, throughout $u, v \geq 0$, $u + v \leq R$,*

$$2\{K_\lambda(u)^2 + K_\lambda(v)^2 + K_\lambda(u+v)^2\} \geq 2(b/B)^2 > 0. \quad (50)$$

Proof. Let $A_X = q \cos X - X \sin X$. For $Z = X + Y$, angle addition gives

$$(X^2 + XY + Y^2 + q^2) \sin X \sin Y = A_X A_Y - X A_X \sin Y - Y A_Y \sin X - q A_Z. \quad (51)$$

Take $X, Y \geq 0$, $Z \leq M$, and put $h = \max\{|A_X|, |A_Y|, |A_Z|\}$. If $h \leq q/2$, then

$$q \leq q|\cos X| + q|\sin X| \leq h + (q + X)|\sin X|,$$

so $|\sin X| \geq q/[2(q + M)]$, and likewise for Y . Taking absolute values in (51) therefore yields

$$\frac{q^4}{4(q + M)^2} \leq h^2 + (M + q)h \leq (M + 3q/2)h.$$

If $h > q/2$, the other alternative in the definition of b applies. In either case $h \geq b > 0$. The identity

$$A_X = q(1 - X^2/a^2)K_\lambda(X/(\pi\lambda))$$

holds also at the removable points by continuity. Since $|q(1 - X^2/a^2)| \leq B$ for $0 \leq X \leq M$, at least one of the three kernel values has absolute value at least b/B . This proves (50). $\square$

The zero-set obstruction itself is already used in [13]. The explicit formula in Lemma 4.1 makes its positive compact gap available for every fixed exponent without a numerical minimum. Its small size is acceptable for the qualitative strict-improvement theorem; Section 10 certifies a more useful gap in one example.

5 Short-interval arithmetic and passage to the sharp density

We record the exact analytic interface to avoid a support or smoothing uniformity assumption. Fix $0 < \lambda < \theta < 1$, put $H = T^\theta$, $L = \log T$, and $I = (T, T + H]$. Wang's Theorem 2.2 states that for every fixed real even $g \in C_c^\infty(\mathbb{R})$ supported in $[-\lambda, \lambda]$,

$$\begin{aligned} W_I(g) &:= \sum_{\gamma, \gamma' \in I} \widehat{g}\left(\frac{i(\rho - \rho')L}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} \\ &= \frac{HL}{2\pi} \left(g(0) + \int_{\mathbb{R}} |\alpha| g(\alpha) d\alpha \right) + O_g(H + T^\lambda L^2). \end{aligned} \quad (52)$$

The constants may depend on fixed λ, θ . This is an unconditional complex-zero formula; RH is not an assumption.

Choose a fixed real even $\eta \in C_c^\infty((-\lambda/2, \lambda/2))$ with $\int \eta^2 = 1$, and write $f = \eta^2$, $K = \widehat{f}$, $g = f * f$. Both g and g'' meet the fixed-test hypotheses. At $z = i(\rho - \rho')L/(2\pi)$, differentiation of the Fourier transform gives

$$\widehat{g''}(z) = -4\pi^2 z^2 K(z)^2 = L^2(\rho - \rho')^2 K(z)^2.$$

The exact identity

$$\left(\widehat{g}(z) - \frac{\widehat{g''}(z)}{4L^2} \right) \frac{4}{4 - (\rho - \rho')^2} = K(z)^2$$

removes the rational weight. Apply (52) separately to the two fixed tests and combine the resulting estimates. As in [5, Lemma 3.1], this gives

$$\sum_{\gamma, \gamma' \in I} K(z_\rho - z_{\rho'})^2 = \mathcal{C}(f) \frac{HL}{2\pi} + O_f(H + T^\lambda L^2), \quad (53)$$

where

$$z_\rho = \frac{i(\rho - 1/2)L}{2\pi}, \quad \mathcal{C}(f) = \int f^2 + \iint |u - v| f(u) f(v) du dv. \quad (54)$$

Indeed, the main term for g'' is $-\int (f')^2 + 2 \int f^2$, finite for the fixed smooth density, and its coefficient is $1/(4L^2)$. A varying scalar coefficient after applying the theorem to two fixed tests does not require a varying-test theorem.

Reflection $\rho \mapsto 1 - \bar{\rho}$ preserves the height interval and multiplicity and maps z_ρ to $\bar{z}_\rho$. Thus the multiset $Z_T = \{z_\rho : \gamma \in I\}$ satisfies Proposition 2.2. Its simple real points are precisely the simple critical zeros. Their real-coordinate span is at most $X_T = HL/(2\pi)$. The short-interval Riemann–von Mangoldt formula gives

$$N(T, H) = X_T + O(H + \log T), \quad X_T/N(T, H) \rightarrow 1. \quad (55)$$

After division by N , the error in (53) is $O_f(1/L + T^{\lambda-\theta}L) = o(1)$. The strict support gap $\lambda < \theta$ is therefore necessary at this stage.

The cosine density (47) is Wang's minimizing density and satisfies

$$\mathcal{C}(f_\lambda) = \frac{\lambda}{2} + \frac{1}{\sqrt{2}} \cot(\lambda/\sqrt{2}) = 2 - c(\lambda). \quad (56)$$

It is not a smooth compactly supported test because its endpoint values do not vanish. Choose even cutoffs $0 \leq \chi_\varepsilon \leq 1$, supported strictly inside $(-\lambda/2, \lambda/2)$ and approaching one in the interior. Normalizing $\eta_\varepsilon = \chi_\varepsilon \sqrt{f_\lambda} / \|\chi_\varepsilon \sqrt{f_\lambda}\|_2$ gives densities $f_\varepsilon = \eta_\varepsilon^2$ converging in both L^1 and L^2 to f_λ . Moreover $f_\lambda \rightarrow f_\theta$ in those spaces as $\lambda \uparrow \theta$, and $\mathcal{C}$ is continuous for these convergences on bounded support [5, Section 4].

For normalized nonnegative densities,

$$\sup_{x \in \mathbb{R}} |\hat{f}(x) - K_\theta(x)| \leq \|f - f_\theta\|_1, \quad |\hat{f}(x)|, |K_\theta(x)| \leq 1. \quad (57)$$

Thus the doubled triple energy changes by at most $12\|f - f_\theta\|_1$. This retains a fixed positive energy margin through the smooth approximation.

Proposition 5.1 (Transfer with a certified limiting gap). *Fix $0 < \theta < 1$, $R > 0$, and $0 < d \leq 1$. Suppose (43) holds for K_θ . Define*

$$c_d = \frac{c(\theta) - 2d/(3R)}{1 - d/3}.$$

Then the simple-critical and distinct proportions in (20) are at least c_d and $(1 + c_d)/2$, respectively.

Proof. First fix $0 < d' < d$. Choose a sequence of fixed smooth densities f_j supported in $(-\lambda_j/2, \lambda_j/2)$, with $\lambda_j < \theta$, converging to f_θ in L^1 and L^2 . By (57), every sufficiently large j has triple energy at least d' . Fix such a j before letting T grow.

Write $s = N_0^s(T, H)$, $N = N(T, H)$, and Q for the pair sum with this fixed kernel. Proposition 2.2 and Lemma 3.2, with actual span at most X_T , imply

$$\left(1 - \frac{d'}{3}\right) \frac{s}{N} \geq 2 - \frac{Q}{N} - \frac{2d'}{3N} - \frac{2d'X_T}{3RN}.$$

Let $T \rightarrow \infty$, using (53) and (55), and then let $j \rightarrow \infty$. The result is $\liminf s/N \geq c_{d'}$.

For the distinct count, the other finite inequality gives

$$\liminf \frac{N^d(T, H)}{N} \geq \frac{1 + c(\theta)}{2} + \frac{d'}{6}(c_{d'} - 2/R) = \frac{1 + c_{d'}}{2}.$$

The equality uses $c_{d'} - c(\theta) = d'(c_{d'} - 2/R)/3$. Finally let $d' \uparrow d$; all scalar expressions are continuous since $d \leq 1 < 3$. No test function changes inside a large- T limit. $\square$

Proof of Theorem 1.5. Lemma 4.1 with $\lambda = \theta$ proves a limiting-kernel gap at least $2(b/B)^2$, and hence at least the positive δ in (18). Apply Proposition 5.1 with $d = \delta$. Its constant equals (19). The gain is strictly positive because $R > 2/c(\theta)$. $\square$

Proof of Theorem 1.6. First fix $0 < \varepsilon' < \varepsilon$, keeping p and k fixed. Choose fixed smooth densities f_j supported inside $(-\lambda_j/2, \lambda_j/2)$, with $\lambda_j < \theta$, converging to f_θ in L^1 and L^2 . The uniform estimate (57) changes the triple energy by at most $12\|f_j - f_\theta\|_1$. Thus every sufficiently large j satisfies the all-gap pressure inequality with the same p and target ε' . Also $k\varepsilon' < 1$.

Put $M = 2k + 1$, $\alpha' = k\varepsilon'/M$ and $\beta = 2kp/M$. For each individual fixed j , Proposition 2.2 and Lemma 3.3 give

$$(1 - \alpha') \frac{s}{N} \geq 2 - \frac{Q}{N} - \beta \frac{X_T}{N} - \frac{\alpha'(M - 1)}{N}.$$

Take $T \rightarrow \infty$ with this test fixed. If $c_j = 2 - \mathcal{C}(f_j)$ and $b_j = (c_j - \beta)/(1 - \alpha')$, then $\liminf s/N \geq b_j$. The stronger finite inequality for distinct points gives

$$2 \liminf \frac{N^d(T, T^\theta)}{N} \geq 1 + c_j + \alpha' b_j - \beta = 1 + b_j.$$

The identity follows from $(1 - \alpha')b_j = c_j - \beta$. It does not use the false general counting shortcut $N^d \geq (N + s)/2$.

Now let $j \rightarrow \infty$, so $c_j \rightarrow c(\theta)$, and finally let $\varepsilon' \uparrow \varepsilon$. The limiting denominator is at least $M - 1 > 0$, including the endpoint $k\varepsilon = 1$. This proves both proportions in Theorem 1.6. No support parameter or smoothing scale varies within a height limit. $\square$

Proof of Corollary 1.7. Lemma 4.1 gives the compact energy bound with $d = \delta$. Nonnegativity outside its triangle gives the all-gap inequality

$$2\{K_\theta(u)^2 + K_\theta(v)^2 + K_\theta(u+v)^2\} + (d/R)(u+v) \geq d.$$

Apply Theorem 1.6 with $\varepsilon = d$ and $p = d/R$. The gain coefficient satisfies

$$\frac{kd}{2k+1-kd} - \frac{d}{3-d} = \frac{d(k-1)}{(2k+1-kd)(3-d)} > 0$$

for $k > 1$. Finally $b \leq q/2$ and $B > q$ in (18), so $2(b/B)^2 < 1/2$ and $0 < \delta = (b/B)^2 < 1/4$. Thus $k = 2$ is always admissible, giving the stronger entire positive curve. $\square$

The same compact-to-pressure conversion works with any stronger certified d , using strict lower targets during smoothing before the final scalar limit. In particular $d = 1/7000$, $R = 21/4$, $k = 7000$ gives (23). Its gain relative to the original triple bound is exactly $(3 - d)/2 = 20999/14000$. This is a ratio of gains above Wang's baseline, not a ratio of the full proportions.

6 Classical odd zeros and the range extension

The positive Gram defect is useful for quantitative pressure improvements. For the qualitative parity theorem we may discard it: the multiplicity excess in the same finite inequality already supplies the missing information when combined with a classical sign-change theorem.

6.1 The exact classical count and interval packing

We use the following fixed consequence of Selberg's theorem, stated as Theorem B on p. 523 of Karatsuba [9]: for the fixed $\alpha = 51/100$ there exist constants $a > 0$ and t_a such that

$$N_{\text{odd}}(t, t^\alpha) \geq at^\alpha \log t \quad (t \geq t_a). \quad (58)$$

The zeros in this count are distinct and of odd order on the critical line. Karatsuba's stronger main theorem, p. 524, gives the same kind of bound for $\alpha = 27/82 + \epsilon$, $0 < \epsilon \leq 0.001$. His final argument in Section 12, p. 536, selects disjoint sign-changing intervals; this confirms the distinct-count convention. We import the classical theorem and its analytic estimates; we do not reprove them. Selberg's original paper was not consulted, so our source for its exact statement is Karatsuba's primary article.

Fix $\alpha < \theta < 1$ and put $H = T^\theta$. Starting at $t_0 = T$, define $t_{j+1} = t_j + t_j^\alpha$ and keep all complete adjacent intervals lying inside $(T, T + H]$. Their total length is $H - O(T^\alpha) = H(1 - o(1))$. Their endpoints are all $T(1 + o(1))$, and hence their logarithms are $\log T + o(1)$. Summing (58) gives

$$N_{\text{odd}}(T, H) \geq (a + o(1))H \log T. \quad (59)$$

If a seed is stated with different open or closed endpoint conventions, at most two distinct points per block must be removed. There are $O(H/T^\alpha + 1)$ blocks, so this loss is $o(H \log T)$ and does not change the conclusion. Counting odd copies instead would not justify this bounded endpoint loss, which is another reason to keep the precise classical convention.

The Riemann–von Mangoldt formula gives $N(T, H) \sim H \log T / (2\pi)$. Choose once and for all a sufficiently small $\kappa > 0$ with $\kappa < 2\pi a$. Then

$$\liminf_{T \rightarrow \infty} \frac{N_{\text{odd}}(T, T^\theta)}{N(T, T^\theta)} \geq \kappa \quad \text{for every fixed } \alpha < \theta < 1. \quad (60)$$

The same κ works because it was chosen from the one fixed seed. Only the eventual height is allowed to depend on θ . Nothing here permits $\theta - \alpha$ to shrink with T .

6.2 Fixed-test transfer, including nonpositive cosine bounds

Fix $\alpha < \theta < 1$ and $0 < \lambda < \theta$. Use the transformed zeta multiset and one fixed smooth probability density $f = \eta^2$ with support in $(-\lambda/2, \lambda/2)$ from the arithmetic transfer above. Its real simple and odd points are precisely those counted by N_0^s and N_{odd} . Equations (53) and (54) give

$$Q/N \longrightarrow \mathcal{C}(f).$$

This is an exact unweighted pair sum followed by an asymptotic limit. The weight removal uses Wang's theorem separately for two fixed test functions, including the differentiated test; it does not assume that individual complex pair terms are nonnegative. Its normalized error is $O_f(1/\log T + T^{\lambda-\theta} \log T) = o(1)$.

Apply Proposition 2.3, Proposition 2.2, $D(G) \geq 0$, $N^d \geq N_{\text{odd}}$, and (60). For this fixed test we obtain

$$\begin{aligned}\liminf N_0^s/N &\geq \max\{0, 2 - \mathcal{C}(f), [2 - \mathcal{C}(f) + 2\kappa]/3\}, \\ \liminf N^d/N &\geq \max\{\kappa, [3 - \mathcal{C}(f)]/2, [7 - 2\mathcal{C}(f) + \kappa]/6\}.\end{aligned}$$

Choose the normalized squares of smooth cutoffs of the cosine square root described earlier, with fixed support parameter λ . After the height limit, let the cutoff converge in L^1 and L^2 to f_λ . Then $\mathcal{C}(f) \rightarrow 2 - c(\lambda)$ by (56). Finally let $\lambda \uparrow \theta$. Continuity gives (14)–(15). This argument is valid when $c(\lambda) \leq 0$; positivity was needed for the pressure gain, not for the pair-sum limit or the finite inequalities.

For completeness, the cosine cost also has a short direct evaluation. Let $J(x) = \int |x - y| f_\lambda(y) dy$. On the support interior, $J'' = 2f_\lambda$ and $f_\lambda'' = -2f_\lambda$. The even function $f_\lambda + J$ is therefore constant. At $b = \lambda/2$ it equals $b + \cot(\lambda/\sqrt{2})/\sqrt{2}$, since $J(b) = b$. Multiplication by f_λ and integration gives

$$\mathcal{C}(f_\lambda) = \frac{\lambda}{2} + \frac{1}{\sqrt{2}} \cot(\lambda/\sqrt{2}) = 2 - c(\lambda).$$

The legal order is thus fixed test, height limit, smooth approximation, and support limit. No test or derivative degree grows with the height.

6.3 Proof of the strict range statements

The definition of θ_0 and the derivative formula require no computed root. To verify $\alpha < \theta_0$, put $a_0 = \alpha/\sqrt{2}$. The elementary bounds $\cos a_0 \geq 1 - a_0^2/2 > 0$ and $0 < \sin a_0/a_0 \leq 1$ imply

$$a_0 \cot a_0 \geq 1 - \alpha^2/4, \quad c(\alpha) \leq 2 - \frac{1}{\alpha} - \frac{\alpha}{4} = -\frac{1801}{20400} < 0.$$

Also $\tan x > x$ on $(0, \pi/2)$ gives $c(1) > 1/2$, and $c'(\theta) > 0$ gives the unique zero. On $[\alpha, \theta_0]$,

$$c'(\theta) = \frac{1}{2} \cot^2(\theta/\sqrt{2}) < \frac{1}{\theta^2} \leq \frac{10000}{2601} < 4. \quad (61)$$

For δ in (16), we have $\alpha < \theta_1 < \theta_0$ and

$$c(\theta_1) = - \int_{\theta_1}^{\theta_0} c'(t) dt > -4\delta \geq -\kappa.$$

The right-hand side of (14) is therefore greater than $\kappa/3$ at θ_1 , and is no smaller at any fixed larger exponent. With δ_{half} in (17), the same argument gives $c(\theta_{\text{half}}) > -\kappa/4$; the separately proved distinct formula yields

$$\frac{3 + 2c(\theta_{\text{half}}) + \kappa}{6} > \frac{1}{2} + \frac{\kappa}{12}.$$

This proves the remaining claims of Theorem 1.4.

At θ_0 itself, (14) gives a lower proportion at least $2\kappa/3 > 0$. The strict extension follows from this positive margin and continuity, without a new pair-correlation support estimate. Its numerical size remains unknown because the imported fixed classical constant has not been evaluated.

7 Localizing the optimized Selberg detector

Pearce-Crump [6] constructs positive-semidefinite vector profiles whose sums of squared mollifiers equal Hardy's real function times a nonnegative weight on the critical line. Sign changes therefore detect distinct odd-multiplicity critical zeros. For a fixed admissible profile $q : [0, 1] \rightarrow \mathbb{R}^J$, write $\beta_{q,U}$ for the analytic half and $U \asymp T^u$ for the mollifier length. The source normalization is

$$\aleph(s)Y_{q,U}(s) = 2 \operatorname{Re} \beta_{q,U}(s) \quad (\operatorname{Re} s = 1/2). \quad (62)$$

The vector-profile diagonal theorem gives

$$S_q(1, U) \leq \frac{C[q] + o(1)}{\log U}. \quad (63)$$

The profile is first regularized by a fixed $\varepsilon > 0$ so that the terminal coefficient needed by the horizontal Jensen estimate is nonzero. The limit $T \rightarrow \infty$ is taken first, followed by $\varepsilon \downarrow 0$; no estimate uniform in ε is used.

Fix $1/2 < \theta < 1$, put $H = T^\theta$, and choose a fixed

$$0 < u < \frac{\theta - 1/2}{2}. \quad (64)$$

Bertrand's postulate supplies a prime U with $T^u \leq U \leq 2T^u$, hence $\log U = u \log T + O(1)$. Use the source's fixed analytic vector on the dyadic block $[T, 2T]$ and apply Littlewood's rectangle argument only on $[T, T + H]$.

The horizontal increment bound applies componentwise. For fixed J , summing the finitely many terms changes only a profile-dependent constant, so it is

$$O_q\left(U \log(2T) \{\log^2(2T) + \eta_U + 1\}\right) = o(H)$$

under (64). The auxiliary zeta and mollifier increments are also finite componentwise sums. On the right auxiliary edge, absolute convergence gives $O_q(1)$. Consequently

$$\frac{\pi x}{2} N_{\text{odd}}(T, H) \geq - \int_T^{T+H} \log |\beta_{q,U}(\delta + it)| dt - H \log 2 - o(H). \quad (65)$$

The diagonal remainder is pointwise and is $O_q(T^{-1/2} U^2 \log^2(2U)) = o(1)$. Pearce-Crump's rational-frequency lemma applies to arbitrary intervals in $[T, 2T]$ and to every fixed mixed component. After division by H , each off-diagonal entry is

$$O_q\left(T^{1/2+2u-\theta} \log T\right) = o(1). \quad (66)$$

There are finitely many entries, so the exponent is unchanged. The approximate-functional-equation error is componentwise $o(1)$ because (64) implies $u < 1/4$. Thus

$$\frac{1}{H} \int_T^{T+H} |\beta_{q,U}(\delta + it)|^2 dt \leq \frac{C[q] + o(1)}{x \log U}. \quad (67)$$

Jensen's inequality, (65), and (67) give

$$N_{\text{odd}}(T, H) \geq \frac{H}{\pi x} \left\{ \log \frac{x \log U}{C[q]} - 2 \log 2 + o(1) \right\}.$$

The right side is optimized at $x = 4eC[q]/\log U$, yielding

$$N_{\text{odd}}(T, H) \geq \frac{H \log U}{4\pi e C[q]}(1 + o(1)). \quad (68)$$

Since $N(T, T^\theta) \sim T^\theta \log T/(2\pi)$, this gives

$$\liminf \frac{N_{\text{odd}}(T, T^\theta)}{N(T, T^\theta)} \geq \frac{u}{2eC[q]}.$$

Taking the supremum over fixed u in (64) proves (3); the rank and mollifier exponent never vary with T .

Apply the pointwise parity inequality $3N_0^s \geq 2N - Q + 2N_{\text{odd}} + D$ with $D \geq 0$. Wang's fixed-test limit gives the $c(\theta)$ contribution from $(2N - Q)/N$, and (3) supplies the odd support. Specializing to the source profile q_6 proves (5). The source prints the interval (4) but not the ten-direction coefficient matrix. The use of C_6 is therefore a normal attributed analytic premise, while the rank-independent localization above is the deduction proved here.

8 Hilbert–parity transfer and the lower threshold

Apply Proposition 2.5 to Wang's scaled zero multiset in $(T, T + T^\theta]$. With

$$q_T = \frac{Q_T}{N(T, T^\theta)}, \quad s_T = \frac{N_0^s(T, T^\theta)}{N(T, T^\theta)}, \quad o_T = \frac{N_{\text{odd}}(T, T^\theta)}{N(T, T^\theta)},$$

division by $N(T, T^\theta)^2$ gives

$$(q_T - s_T)(1 - o_T) \geq 2(1 - s_T)^2. \quad (69)$$

No product of liminf values is used. Along a subsequence realizing the lower limit of s_T , Wang supplies an eventual fixed-test upper bound for q_T and Theorem 1.1 supplies $o_T \geq k_6(\theta) - \varepsilon$. Insert both bounds before sending ε to zero. The smaller root of the resulting quadratic is

$$R(A, b) = \frac{4 - b - \sqrt{b\{b + 8(A - 1)\}}}{4}. \quad (70)$$

Use Wang's legal smoothing order. Since the limiting pair term is $A = 2 - c(\theta)$ and $b = 1 - k_6(\theta)$, (70) becomes $h_6(\theta)$ and proves (6).

The new term is positive exactly when

$$F_6(\theta) := c(\theta) + (2 - c(\theta))k_6(\theta) > 0. \quad (71)$$

On $(1/2, 1)$,

$$F_6'(\theta) = c'(\theta)(1 - k_6(\theta)) + (2 - c(\theta))k_6'(\theta) > 0,$$

so the onset is unique. Exact directed rational enclosures prove opposite signs at 0.5458837 and 0.5458838, giving (7). Repeating the same calculation with the fully printed C_3 profile places its root in $(0.5458846, 0.5458847)$. At the separating exponent 0.545884, the rank-three term is negative and (8) is positive.

At $\theta = 0.5459$, exact evaluation gives (9) and

$$h_6(0.5459) - h_3(0.5459) > 9.2635430617773560 \times 10^{-7}.$$

Here h_3 denotes the same root formula with $k_3 = (\theta - 1/2)/(4eC_3)$. The improvement is about 5.5 percent relative to the old lower bound at this point.

The improvement cannot be recovered from the scalar pair and odd-support headlines alone. A normalized mixture of real triple and double points can satisfy the scalar constraints with zero simple support. The product (37) rules it out at Wang's actual pair level because it retains the first Hilbert dimension before compression.

9 Spectral-defect parity transfer

Normalize Proposition 2.6 by N^2 and write $d_T = D(G_T)/N$. The finite inequality is

$$(q_T - s_T - d_T)(1 - o_T) \geq 2(1 - s_T)^2. \quad (72)$$

Both factors on the left are nonnegative. Suppose an all-gap pressure certificate (21) is available with an integer $\ell \geq 1$ and $\ell\varepsilon \leq 1$. Put

$$M = 2\ell + 1, \quad \alpha = \frac{\ell\varepsilon}{M}, \quad \beta = \frac{2\ell p}{M}.$$

The disjoint-frame and span argument used in Theorem 1.6 gives

$$\liminf_{T \rightarrow \infty} d_T \geq \alpha s - \beta, \quad s = \liminf_{T \rightarrow \infty} s_T. \quad (73)$$

Choose a subsequence on which $s_T \rightarrow s$. For every positive η , the rank-six odd-support theorem and (73) give eventually $o_T \geq k_6(\theta) - \eta$ and $d_T \geq \alpha s - \beta - \eta$. Wang's fixed-test theorem gives $q_T \rightarrow q_\theta$ in the legal smoothing order. Insert the eventual bounds into the nonnegative factors of (72) before sending η to zero. This yields

$$2(1 - s)^2 \leq (1 - k_6(\theta))\{q_\theta + \beta - (1 + \alpha)s\}. \quad (74)$$

The smaller root is (10).

For the universal choice in Theorem 1.3, use the analytic triangle estimate behind (18). Put $h = h_6(\theta) > 0$, $R = 11/(5h)$, and

$$\varepsilon = d_\theta = (b/B)^2, \quad p = d_\theta/R, \quad \ell = 2.$$

The proof of Corollary 1.7 gives $d_\theta < 1/4$, so this choice is admissible, and

$$\alpha_\theta = \frac{2d_\theta}{5}, \quad \beta_\theta = \frac{4d_\theta h}{11}, \quad \alpha_\theta h - \beta_\theta = \frac{2d_\theta h}{55} > 0.$$

The ratio 11/5 was frozen before the authoritative computation. It is a simple rational near the large-radius asymptotic optimum; no global optimality claim is needed.

Let F be the left side minus the right side of (74). The old root h makes the same equation with $\alpha = \beta = 0$ an equality, so

$$F(h) = (1 - k_6(\theta))(\alpha_\theta h - \beta_\theta) > 0.$$

As in EXP-007, $F(1) < 0$, the larger root exceeds one, and on the interval from h to the smaller new root H_6 one has $-4 \leq F' < 0$. The mean value theorem gives

$$H_6 - h \geq \frac{(1 - k_6(\theta))(\alpha_\theta h - \beta_\theta)}{4} > 0.$$

This proves Theorem 1.3 before numerical evaluation.

The construction does not change the onset. At $h_6(\theta) = 0$, the radius $11/(5h_6(\theta))$ is unavailable and the simple-real Gram defect may vanish. An onset improvement requires a stronger independent input, which the source-certified C_6 supplies, or a statistic that remains informative when $S = 0$.

10 The compact certificate

For $\theta = 3/4$ and $R = 21/4$, a finite interval computation proves

$$2\{K_{3/4}(u)^2 + K_{3/4}(v)^2 + K_{3/4}(u+v)^2\} \geq \frac{1}{7000} \quad (u, v \geq 0, u+v \leq 21/4). \quad (75)$$

The certificate is a complete binary partition of the square $[0, R]^2$. Every endpoint and threshold is rational. A box is either subdivided, proved wholly outside the triangle, or accepted using an energy lower bound valid on the entire box. The final tree contains 48,761 nodes, 24,252 accepted energy leaves, 129 outside leaves, and no unresolved boxes (computation EXP-002 [19]).

For completeness, the analytic enclosure used at each leaf is simple. Since f_θ is a probability density on $[-\theta/2, \theta/2]$,

$$|K'_\theta(x)| \leq 2\pi \int |u| f_\theta(u) du \leq \pi\theta. \quad (76)$$

On a real interval with center x_0 and radius r , an outward interval evaluation of $K_\theta(x_0)$ therefore supplies the valid lower bound

$$|K_\theta(x)| \geq \max\{0, \text{lower}(|K_\theta(x_0)|) - \pi\theta r\}.$$

Apply this separately to the box ranges for u , v , and $u+v$, square the nonnegative lower bounds, and double their sum. Every acceptance comparison requires the entire lower enclosure to exceed the rational threshold. An outside box is discarded only if the sum of its two lower coordinates is strictly greater than R , so the triangle boundary is retained.

Discovery used Arb through `python-flint==0.9.0` at 160-bit precision and the removable-singularity-free formula (49). Each accepted leaf was replayed at 256 bits with a separately derived Taylor evaluator. For real x ,

$$\text{sinc}(x) = \sum_{j=0}^{95} \frac{(-1)^j x^{2j}}{(2j+1)!} + \mathcal{R}(x), \quad |\mathcal{R}(x)| \leq \frac{|x|^{192}}{193!}. \quad (77)$$

The remainder follows by applying the real sine Taylor theorem through degree 192 and dividing by x , with continuous extension at zero. Arb encloses both the polynomial and its remainder. This replay is an independent kernel evaluator; it shares the partition reconstruction, Lipschitz argument, and Arb library with the first computation. It is not a second independent full verifier or a Lean proof.

Proof of Corollary 1.8. Use the finite certificate (75) in Proposition 5.1, with $d = 1/7000$ and $R = 21/4$. The full threshold d is available because the proposition first uses every $d' < d$ through smoothing and then passes to d . Substitution gives the stated constants. The outward rational enclosures for their values and positive difference are recorded with the certificate. $\square$

The certificate's role is quantitative. Even without it, Theorem 1.5 proves a strict improvement for every fixed $\theta > \theta_0$ using its explicit analytic gap. Conversely, the finite numerical inequality alone does not prove a theorem about zeta zeros without the finite operator and fixed-test analytic transfer.

Proportion at $\theta = 3/4$	Wang baseline	This certificate
Simple and critical	0.4190750129754243...	0.4190768284253039...
Distinct	0.7095375064877121...	0.7095384142126519...

Table 1: All denominators count nontrivial zeros with multiplicity in $(T, T + T^{3/4}]$. The entries are proportions, not percentages.

11 Conclusions and further questions

The spectral-defect product (40) is the strongest finite result of this paper. The rank-independent localization is the new analytic interface: it permits the source-certified six-square detector to replace the fully printed rank-three profile. Together they move the unique positivity onset to $(0.5458837, 0.5458838)$. The optimized spectral transfer then strictly strengthens every positive point of that new curve. The finite coefficient two is sharp, and the proof separates the deductions from the attributed rank-trace and six-square inputs. None of these results proves the Riemann hypothesis.

The next analytic task is to reduce the $T^{1/2}U^2/H$ off-diagonal barrier or independently reconstruct and improve the six-square profile. The published C_6 interval is usable as a cited theorem, but the missing coefficient matrix prevents a local reconstruction. A broader semidefinite search with a printed rational profile would make the new onset fully replayable from first principles. Another direction is to find a statistic that survives when the simple support is zero. Signed Krein-space inertia by itself is insufficient because a conjugate pair can approach the critical line with arbitrarily small negative eigenvalue. Semidefinite optimization can improve kernels only within an unconditional Fourier-support range; beyond it, the missing arithmetic input remains. Probabilistic laws for projected ordinates and zero-density concentration for finite Dirichlet polynomials expose useful interfaces but do not control zeta’s horizontal displacement. Any next statistic must survive the fixed-test limiting process and have a complete analytic interface, not only a finite optimizer or analogy.

Declaration of generative AI use. Generative AI tools were used for derivation, source research, review, and implementation. The author reviewed and verified all content and takes full responsibility for the work.

A Reproducibility, provenance, and scope

The repository github.com/fsantibanezleal/CAOS_RESEARCH archives each computation cited here under an identifier EXP-nnn, in the folder whose name starts with that identifier under `problems/number-theory/riemann-hypothesis/experiments/`. Each such folder contains the statement tested, the proof, the code, its outputs, and the verification of the result. The folder of computation EXP-002 [19] contains the mathematical proof, an independent verification of the proof, the interval runner, the canonical partition tree, and rational enclosures for the constants. A deterministic floating grid selected candidate radii; its sampled values were never used as certified lower bounds. The final certification used CPU arithmetic; no GPU was necessary.

From the repository root, with the problem’s pinned requirements installed, a fresh output directory preserves the canonical artifacts:

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-002-short-interval-stability/run.py \
  --output-dir tmp/riemann-replay --radius 21/4 --threshold 1/7000
pytest tests/test_riemann_certificates.py
```

The displayed backslashes indicate line continuation for readability; use one command line on shells with different continuation syntax. The runner fails with an exact resumable checkpoint if its time or node limit is reached while boxes remain. It does not report a certificate in that state.

The tree replay checks its hash, reconstructs every box, verifies all accepted leaves, and rejects missing, trailing, unknown, or miscounted nodes. A hash protects accidental corruption; geometric replay is still required when a modified tree has a recomputed hash. Negative controls confirmed the rejection of false whole-square acceptance, a false outside-domain claim, truncation, appended data, altered counts, an unresolved flag, an invalid exponent, and an inflated threshold. The focused tests include one complete 256-bit certificate replay. These are implementation checks, not formal verification of the entire analytic proof.

The canonical preorder tree has SHA-256 digest

78829635026aeacd69c02213cc899b5029ef2566bbfaf1c86b757cc5f92ffe55

B The two-variable pressure certificate

For the rational parameters (24), pressure alone proves (21) when $u + v \geq R_{\text{cut}} = \varepsilon/p = 443239/80000$. A finite interval computation certifies the remaining closed triangle. It reconstructs a complete binary partition of the square $[0, R_{\text{cut}}]^2$. Each leaf is either accepted by an outward-rounded energy-plus-pressure bound on its entire rectangle, or by the exact inequality $p(u_0 + v_0) \geq \varepsilon$ at its lower corner. Equality at the cutoff is permitted. Outside the square, pressure alone suffices.

The energy enclosure is the one justified in (76); the pressure contribution on a box is bounded below by $p(u_0 + v_0)$. The certificate has 16,797 nodes, 8,351 validated energy-plus-pressure leaves, 48 pressure-only leaves, and zero unresolved boxes. Construction uses Arb at 160 bits. Every accepted leaf is replayed at 256 bits with the separately derived sinc-Taylor evaluator. These evaluators share Arb, geometric subdivision, and the Lipschitz estimate; this is not a wholly independent verifier or a formal proof of the analytic theorem.

Both bounds of Corollary 1.9 are established in computation EXP-003 [20]. For C_A in (23), the computation replays the 48,761-node compact certificate of Section 10 completely at 256 bits, in about 26.9 seconds, and checks two symbolic identities and 328 small-index incidence/boundary cases. For C_B , it constructs the two-variable certificate described above and replays every accepted leaf, in about 5.2 seconds in total. The gain of C_B above Wang’s bound is

$$C_B - c(3/4) = 0.000012875194687394224537573076\dots,$$

and it exceeds $5/4$ of the gain of C_A :

$$(C_B - c(3/4)) - \frac{5}{4}(C_A - c(3/4)) = 0.000009471388256765351202430110\dots > 0.$$

Together with $k\varepsilon = 999947184/10^9 < 1$, the finite certificate and Theorem 1.6 prove the statements on C_B in Corollary 1.9. Floating-point sampling was used only to choose the rational

parameters (24), which are not claimed to be optimal; every inequality in the certificates is certified in exact rational or ball arithmetic.

Construction checkpoints are candidate proof state; a completed certificate is accepted only after the final arithmetic replay, and a replay restarted from a checkpoint checks the checkpoint identity and revalidates its entire prefix. The focused tests include tampering controls: a forged completed prefix with recomputed hashes is rejected by the leaf arithmetic.

The canonical result, the pressure proof tree, provenance hashes, and the exact rational endpoints are archived in the EXP-003 record [20]. Mathematical artifacts use deterministic UTF-8/LF bytes; elapsed-time logs are separate metadata. To reproduce with the repository virtual environment, run each command with a fresh output directory. The first command recomputes C_A , the second tests checkpointing and restart, and the third runs the focused tests:

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-003-odd-frame-pressure/run.py --stage=A \
    --output-dir tmp/riemann-a
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-003-odd-frame-pressure/run.py --stage=smoke \
    --output-dir tmp/riemann-smoke
python -m pytest tests/test_riemann_pressure.py
```

The full command line for the two-variable certificate is in the EXP-003 record [20]. The displayed backslashes are typographical line continuations; combine each command on shells that use another continuation convention.

C Parity verification and its analytic boundary

The EXP-004 record [21] separates the exact certificates (34)–(35), universal atom formulas, finite regression censuses, the scalar relaxation (36), sharpness examples, and the check of the analytic sources. No interval optimization, zeta-zero sampling, or GPU computation is needed for this route. In particular, the classical κ and the new numerical exponent are deliberately unevaluated fields, not outputs of a finite census.

Independent checks of the proof of the qualitative range theorem, within the stated analytic inputs, cover the signed finite operator, multiplicity charges, distinct-odd convention, interval packing, exact deweighting, and legal limiting sequence. A separate [proof-review file](#) lists the SHA-256 hashes of the proof, the proof checks, the statement tested, the computed result, and the summary of the outcome.

The arithmetic run took about 8.04 seconds. It verified two symbolic residual identities and nonnegative arbitrary-multiplicity atom families, supplemented by 24 single-atom cases through multiplicity 12. The full census consists of 19,683 count vectors in $\{0, 1, 2\}^9$, with 59,049 evaluations at three nonnegative slack values. All 42 rational scalar-relaxation cases have matching feasible primal and dual witnesses. All 36 finite sinc simple/double-support examples attain equality, while a real triple point rejects the false distinct half-sum with residual -1 . The exact derivative identity and rational comparisons in (61) passed.

The focused software tests check tampered counts, the exact bytes of the committed source files, preservation of an incomplete census prefix, input rejection, and protection of existing outputs against overwriting. In two independent recomputations, every stored census row was parsed with direct formulas without importing the runner; all count vectors, residuals, linked artifact hashes, and intentionally null numerical constants matched. The original run outputs are archived

unchanged in the experiment record. Operational timing is kept separate from deterministic mathematical bytes.

The canonical result has SHA-256 digest

bc9e28ca1792657e24da77268e6e2108c3f52684b7e16504f07b32cfce014d6e

To reproduce the finite computation in a fresh directory, use

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-004-parity-density-transfer/run.py \
    --output tmp/riemann-parity-replay
pytest tests/test_riemann_parity.py
```

The displayed backslashes are typographical line continuations. The run verifies the algebra and the finite controls above in exact arithmetic. The universal atom argument, Selberg’s theorem, interval packing, Wang’s fixed-test transfer, and the approximation limits supply the unbounded quantifiers. A finite computation cannot establish those analytic statements by enumeration. The census slack samples are not claimed to be geometric realizations of zeta zeros, and the scalar optimum concerns only the stated relaxation.

D Local Selberg verification

The EXP-005 computation [22] verifies the finite constants in Theorem 1.1. Its primary evaluator uses exact rational Taylor enclosures for $\sqrt{2}$, e , sine, and cosine. A separate 100-decimal interval implementation must lie inside those rational bounds. The run also checks the exact source bytes for Pearce-Crump’s PDF and TeX archive, the strict localization exponent, a negative point at $\theta = 0.5459$, and rejection of the zero-margin mollifier boundary.

The canonical result has SHA-256 digest

3f0ca476c0e2fe688e4e4f43fc11861d9491b3066d067e46bf88d1a441c696a5

and passed all fourteen checks. Its fixed parameters are $\theta = 273/500$, $u = 2299/100000$, and source radius 10^{-17} for C_3 . To reproduce the computation in a fresh directory, use

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-005-local-selberg-transfer/run.py \
    --output-dir tmp/riemann-exp005-replay --budget-seconds 60
pytest tests/test_riemann_local_selberg.py
```

The finite computation does not establish the all-height localization. A separate proof check follows the source argument term by term and verifies the arbitrary-subinterval quantifier and square-root cutoff. It also checks the diagonal and approximate functional equation remainders, horizontal bounds, right-edge integrals, factor-two normalization, regularization order, prime mollifier selection, endpoint convention, and the match between distinct sign changes and the parity count. The recent analytic preprints remain imported mathematical inputs rather than machine-formalized theorems.

E Hilbert–parity verification

The EXP-006 record [23] separates the universal proof from its finite and transcendental checks. The canonical run began from clean commit `0d736fa22ce7e833200381a32e8cc89f77c660e8`; its result has SHA-256

`82c4761b5c97011ff86cdd379d647ad0f94643a7eb8324a4a09aa37f58848bbf`

and records PASS in 0.953 seconds. The exact census checked 18,479 multiplicity profiles: 135 equality cases, 18,340 strict positive-dimension cases, and four empty-dimension cases. The census is diagnostic; the proof of Proposition 2.5 supplies the unbounded quantifiers.

The numerical certificate uses exact fractions, directed Taylor enclosures for e , $\sqrt{2}$, sine, and cosine, and an exact rational square-root enclosure. A separate 100-decimal interval evaluation lies inside every reported rational enclosure. It also verifies the scalar zero-simple witness, endpoint signs in (7), the negative old bound, and the positive new bound in (9). Seven focused tests include tampered source, weakened-bound, and invalid-branch controls. Reproduce it with

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-006-hilbert-parity-compression/run.py \
    --output-dir tmp/riemann-exp006-replay --budget-seconds 120
pytest tests/test_riemann_hilbert_parity.py
```

The earlier broad-bracket and weaker-transfer runs remain archived. CPU arithmetic sufficed; no GPU workload was justified.

F Spectral-defect parity verification

The EXP-007 record [24] separates the universal operator proof, the limiting argument, and the finite arithmetic. The canonical run began from the clean commit

`d63111ffa8a348c51eb4fd06f5a1e70a51211576`

Its result has SHA-256

`ad635c5b60c4bcae63199fb54a7979a02206ce0ee572853b2df13933dafc320c`

and records PASS in 94.719 seconds under a 180-second CPU budget.

The exact spectral census checked 652,260 rational spectra for five values of t , including 3,447,704 equality eigenvalues and 401,436 strict eigenvalues. The multiplicity census checked 18,479 profiles, 55,425 defect trials, 270 equality trials, 55,155 strict trials, and four empty-dimension profiles. These censuses test branch and bookkeeping identities; Proposition 2.6 supplies the unbounded quantifiers.

At $\theta = 0.5459$, directed rational arithmetic proves

$$d_\theta > 1.6416710022140455631116821409474 \times 10^{-64}$$

and the correlated gain (13). Direct subtraction of two independently enclosed roots is too wide to certify that tiny sign. The proof instead uses (12), and the artifact reports both the naive enclosure and the positive correlated bound. A separate 100-decimal `mpmath.iv` implementation overlaps every shared rational enclosure.

The historical pressure parameters at $\theta = 3/4$ are a negative control: they lower the coupled-product root by about 1.6909×10^{-5} , while their direct pressure-only bound remains stronger. The canonical result records this classification rather than assuming a favorable sign. Four focused tests and the combined 81-test EXP-003 through EXP-007 suite pass. Reproduce the focused record with

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-007-spectral-defect-parity/run.py \
    --output-dir tmp/riemann-exp007-replay --budget-seconds 180
python -m pytest -q tests/test_riemann_spectral_defect_parity.py
```

Two failed attempts remain archived: one corrected the consistency relation between independently rounded enclosures, and one exposed the negative sensitivity control. CPU arithmetic was sufficient; no GPU workload was justified.

G Rank-six local-transfer verification

The EXP-008 record [25] audits the finite-rank localization separately from the imported six-square constant. Its canonical run began from the clean commit

5f4885c70188a7126f0b47ea954e589a7583b358

Its result has SHA-256

56db06037e0b36ed519919c352fb0a5419e2b6ba29629d65948f399c1d3bf4ba

and records PASS in 64.406 seconds under a 120-second CPU budget.

Directed rational arithmetic proves the disjoint onset brackets

$$0.5458837 < \theta_6 < 0.5458838, \quad 0.5458846 < \theta_3 < 0.5458847,$$

the separating sign at 0.545884, (9), and

$$h_6(0.5459) - h_3(0.5459) > 9.2635430617773560 \times 10^{-7}.$$

For the spectral companion, it proves $d_\theta > 1.1072486681807148 \times 10^{-61}$ and the correlated gain (13). A separate 100-decimal `mpmath.iv` implementation overlaps every shared enclosure. Four focused calculation tests plus the canonical artifact-binding check pass. Reproduce the record with

```
python problems/number-theory/riemann-hypothesis/experiments/\
EXP-008-rank-six-local-transfer/run.py \
    --output-dir tmp/riemann-exp008-replay --budget-seconds 120
python -m pytest -q tests/test_riemann_rank_six_local.py
```

The six-square coefficient matrix is not printed in the source. The audit therefore validates the localization and scalar consequences relative to the attributed C_6 input; it does not claim an independent reconstruction of that profile. CPU interval arithmetic sufficed, so no GPU workload was justified.

H Source comparison and limitations

The source comparison was updated on 2026-09-20. It used the full Wang v1 manuscript, Lamzouri v2, the current Alpöge–Furman paper, Pearce-Crump v1, and pinned snapshots of the directly relevant global stability, pressure-frame, Bellman-coboundary, and mixed-certificate repositories. Queries combining the Wang identifier, short intervals, pressure, stability, Gram defects, and exact numerical signatures found no prior public statement of these short-interval refinements. The exact odd-frame schedule was not located, but is a simple instance of known global pair capacities and pressure methods. The globally asserted spectrum-energy envelope is excluded from new-theorem claims. This supports the scoped contribution claimed here, without establishing exhaustive priority or detecting unpublished concurrent work. The known finite stability inequality and cosine-root mechanism are explicitly credited to the repository identifier `ainta`; no personal author identity is inferred from that identifier.

For the parity theorem, the source comparison also covered Karatsuba’s complete primary paper, Steuding’s archived dissertation and published abstract, and *Short mollifiers of the Riemann zeta-function* [12]. The short-mollifier result concerns critical-zero density using higher derivatives; it supplies no automatic simple-zero theorem. Steuding’s stated localized theorem uses the first derivative. His proof suggests a route to fixed higher derivatives, but a uniform two-shift extension, the detector’s exact counting convention, and a certified explicit seed would all require additional proof. None is assumed here. Likewise, weighted all-critical Gram defects and negative off-line spectral mass are separate research directions, not premises of the parity theorem.

A bounded search of primary literature and the pinned contributor sources did not locate the exact Hilbert–parity product (37), its quadratic short-interval transfer, the earlier short-interval parity combination, or the localization of Pearce-Crump’s detector. The exact-phrase search included the AxiomMath formalization snapshot and the dated successor repositories. The finite accounting is already a consequence of known general multiplicity inequalities; the contribution is its proved asymptotic consequence with classical and explicit odd-zero inputs. This is not an assertion of exhaustive priority or a replacement for specialist peer review.

The EXP-007 comparison also inspected Anthropic’s parameterized `RankTraceMult.lean`, its abstract tightness result, and the teal-sea eigenbasis bridge [3, 4]. Those sources already contain the arbitrary parameter and the $t = 2$ spectral interpretation. They are credited as prior work. The scoped new deduction is the simultaneous retention of $D(G)$ and $N - O$ in (40), together with the coupled short-interval consequence (11). The bounded search did not locate that product or consequence; it cannot guarantee absolute priority.

Two post-cutoff preprints were also evaluated. Dubon’s concentration theorem concerns zeros of finite Dirichlet polynomials [7]; transferring it to zeros of the completed zeta function would require a separate approximation theorem. Najnudel and Nikeghbali prove an unconditional Cauchy law for projected ordinates [8], while their stronger logarithmic-derivative comparison requires small total horizontal displacement. Neither result is a premise of the present theorem, and neither supplies the missing onset-changing statistic.

The analytic inputs include the recent Wang and Pearce-Crump preprints and the classical sign-change theorems, not an end-to-end machine formalization. The mathematical and numerical checks can still contain errors, and some numerical evaluators share arithmetic components. The explicit local, Hilbert–parity, and defect–parity theorems do not give an effective starting height, uniformity for moving exponents, a global-record proportion, universal simplicity, or elimination of all off-line zeros. The finite product is elementary once the attributed coefficient inequality is granted, but the short-interval consequence still depends on the recent Wang and Pearce-Crump

preprints.

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